

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1703

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks: 20

Annexure A

Descriptive Written Test

Code of Conduct:

I E. RAGUL / 192511032 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

E. Ragul

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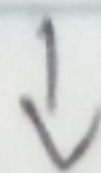
Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

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SMART HOME

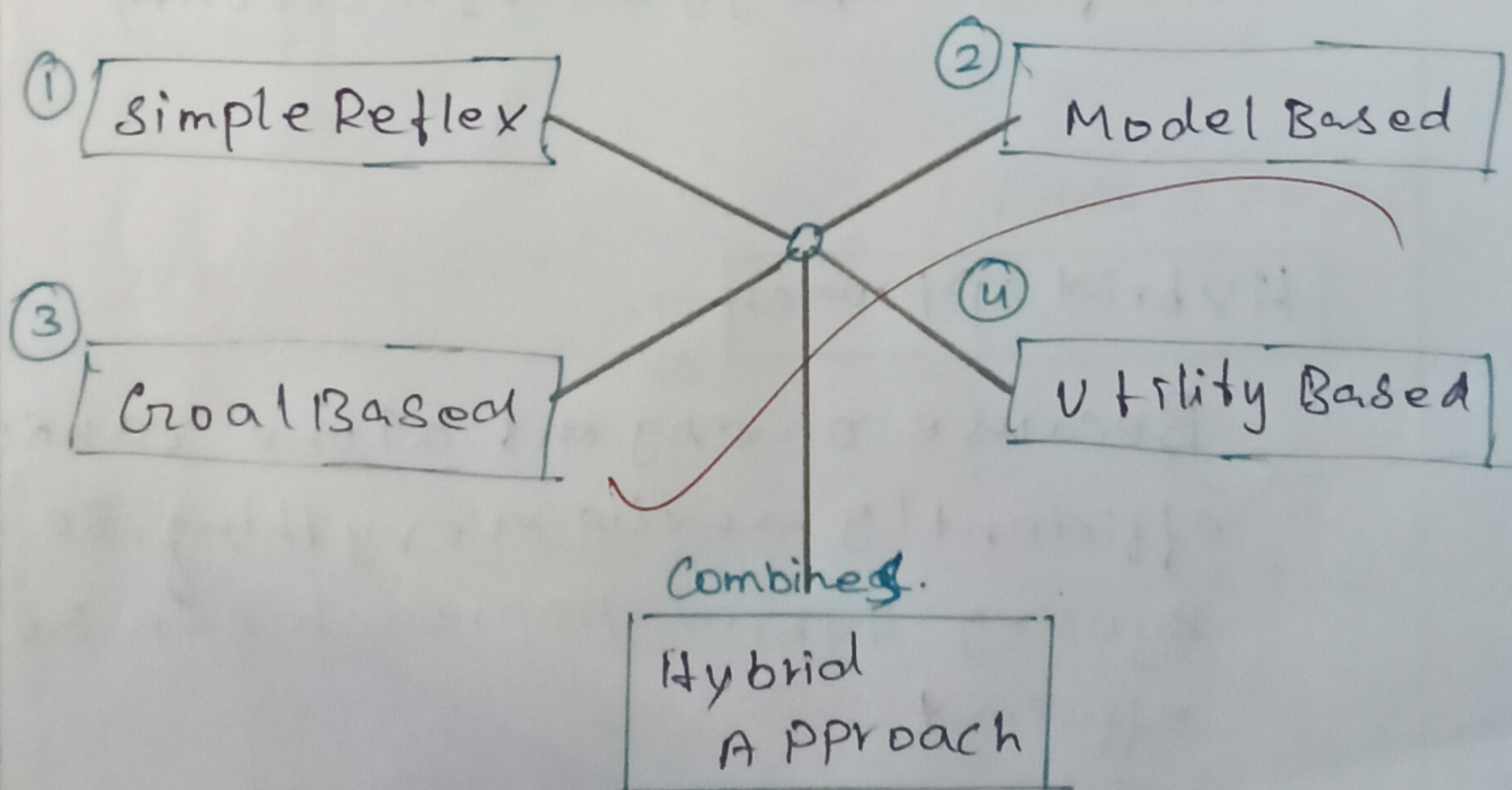


Automatically
Make Decisions

For example:

- A person enters a room: The light should turn ON
- The temperature becomes high: The AC should turn ON
- No body is in the room: the light should become active

Four Types of Agent



• Simple Reflex Agent

* Make decisions based only on the current situation

Ex: Motion detected $\rightarrow$ Turn ON Light

• Model Based Agent

* Uses the current situation + previous information

Ex: The system remembers that a person is inside the room and keeps the light ON

• Goal Based Agent

* Takes actions to achieve a specific goal

Ex: Goal = Save energy Turn off the room's light

• Utility Based Agent

* Chooses the action that gives the Best overall result

Ex: Comfort + Energy Saving + Security

Hybrid approach $\downarrow$

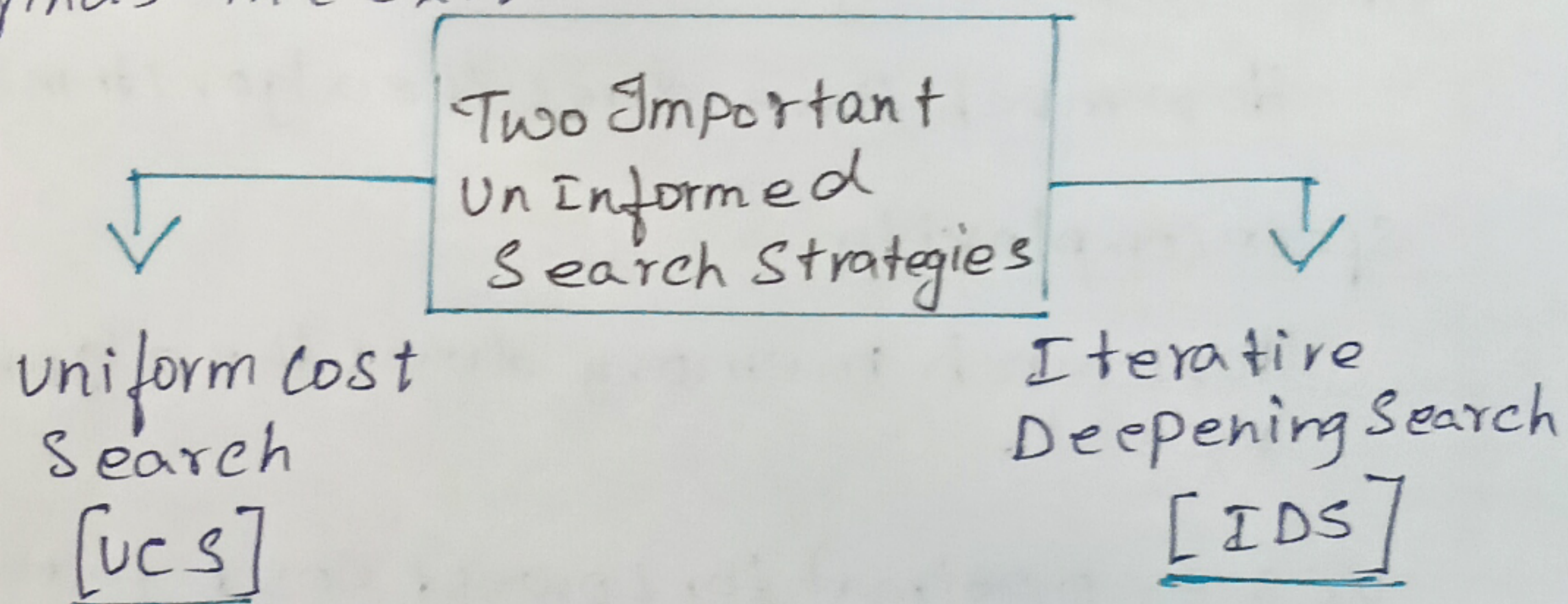
Because one agent alone cannot efficiently handle everything. So By using Hybrid Approach can be efficient.

It will be

②

The robot is Trying to escape from an unknown Maze - complex Network of Paths

→ The robot does not know the maze layout
It has to explore different paths until it finds the exit



Uniform cost search expands the node having the lowest Path cost

It uses priority queue

Ex:
Path A → cost = 10
Path B → cost = 50

UCS use Path B because
It has the lower cost

IDS repeatedly Performs DFS with increasing depth

It uses Stack

Ex:
Depth 0 →
Depth 1 → Depth 2
Depth 3

It continues until
finds the exit.

UCS is useful when different paths have different costs

EDS is useful when robot does not know how deep the solution is

Time complexity

How much time does the algorithm need?

Space complexity

How much memory does the algorithm need?

UCS → optimal for lowest cost paths, but

uses more memory

EDS → uses less memory, but repeats searches and may take more time.

→ Simple Reflex Agent

↳ Reacts to the current situation

→ Model Based Agent

↳ Remembers information about the world

→ Goal Based Agent

↳ Has the goal of reaching the exit

UCS can find the least cost path. While the utility-based agent can evaluate factors such as energy, distance, and safety.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks:
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand